Restaurant Price Optimization

A Product Analytics Experimentation Case Study

Recommended treatment: D - Anchor + Combo

Primary metric: Contribution profit per session

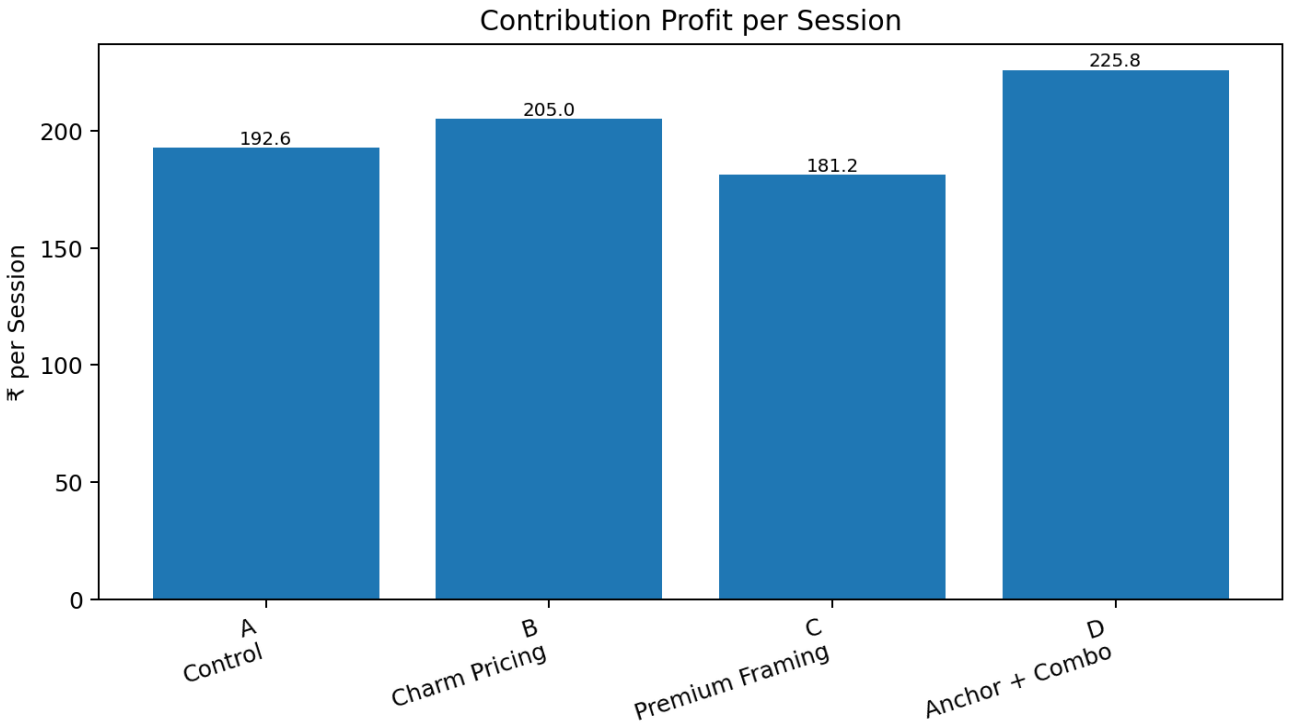
Modelled impact: Rs. 1,659,531 incremental contribution profit per month at 50,000 sessions

Data integrity note: Genuine UCI Restaurant & Consumer Data was used for consumer profiles, restaurant attributes and observed ratings. Treatment assignment and commercial outcomes are simulated because suitable public production A/B-test data was not available. All simulated fields are labelled.

1. Executive Summary

The project evaluates four digital menu-pricing experiences: control, charm pricing, premium framing and anchor-plus-combo. The decision framework prioritizes contribution profit per session while monitoring conversion, order value and discount dependence.

Metric	Control	Winner	Lift
Conversion	49.8%	53.8%	8.0%
Revenue/session	Rs. 314.02	Rs. 366.84	16.8%
Profit/session	Rs. 192.64	Rs. 225.83	17.2%



2. Problem Statement and Product Objective

The restaurant needs to determine whether pricing presentation can increase profitability without reducing customer trust or relying on blanket discounts. The product objective is to select a menu experience for a live pilot using a pre-defined metric hierarchy.

Primary metric

Contribution profit per session, because it combines conversion, realized basket value and variable fulfilment costs into one decision metric.

Secondary metrics and guardrails

Conversion rate, average order value, revenue per session and units per order. A live test should also monitor cancellations, refunds, complaints, repeat purchase and margin dilution.

3. Genuine Data Source and Simulation Method

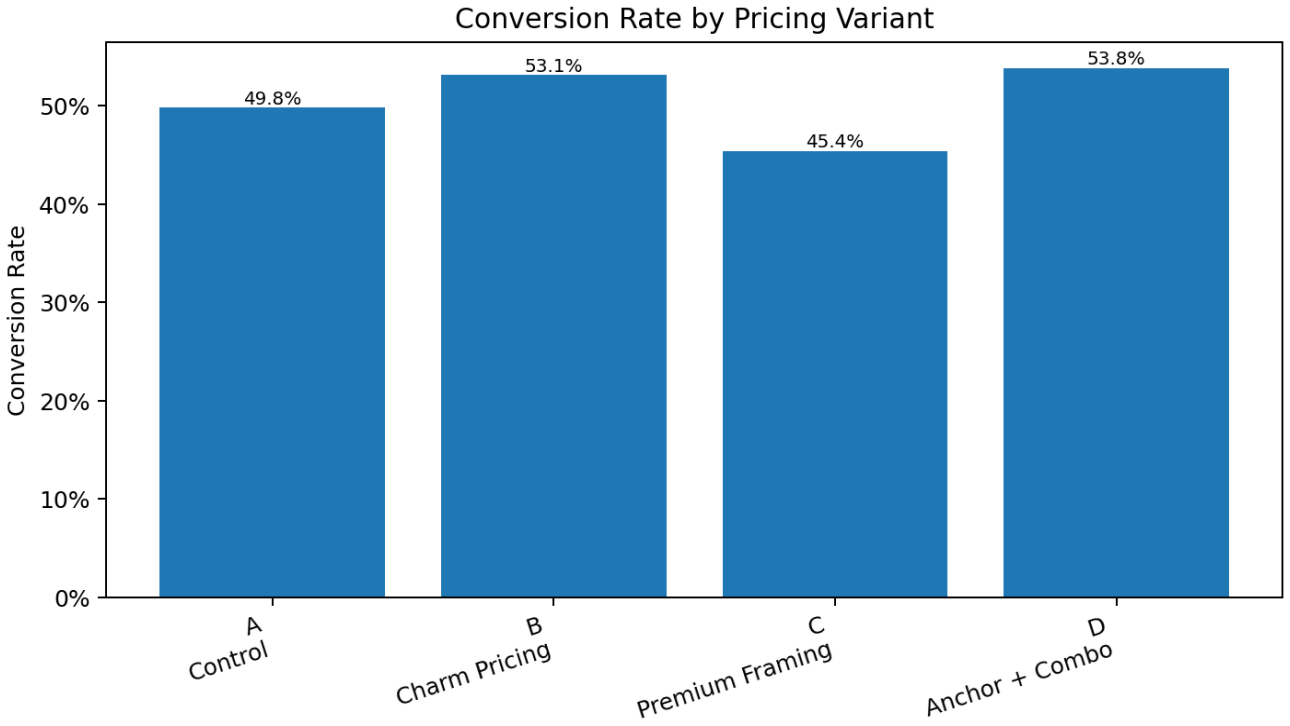
The project uses the UCI Restaurant & Consumer Data dataset (DOI 10.24432/C5DP41, CC BY 4.0), containing 1,161 observed ratings, 138 consumer profiles and 130 restaurant records. Profiles include budget and activity; ratings include overall, food and service scores.

Observed user ratings were converted into baseline purchase-propensity signals. Budget, activity segment, order channel and weekend context were incorporated. Each pricing variant applied transparent effects to conversion, basket size, price realization and discounting.

The numerical uplift is therefore modelled rather than achieved. This distinction must be stated in interviews and on portfolio pages.

4. Experiment Design

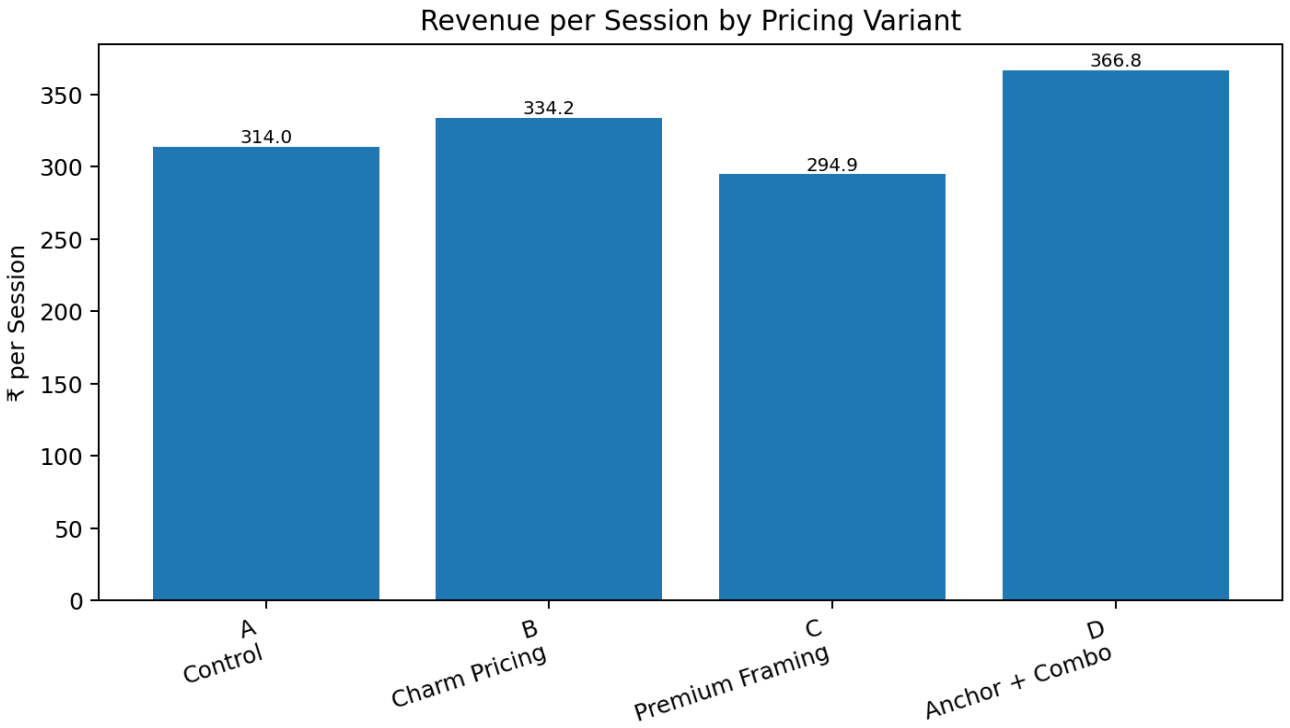
Variant	Experience	Product mechanism
A - Control	Rounded prices; standard descriptions	Causal benchmark
B - Charm Pricing	Prices ending in 9	Lower perceived price for budget-sensitive users
C - Premium Framing	Premium quality cues and higher reference value	Increase willingness to pay; possible conversion trade-off
D - Anchor + Combo	Premium anchor plus value combo	Improve perceived value and increase basket size



5. Results and Statistical Validation

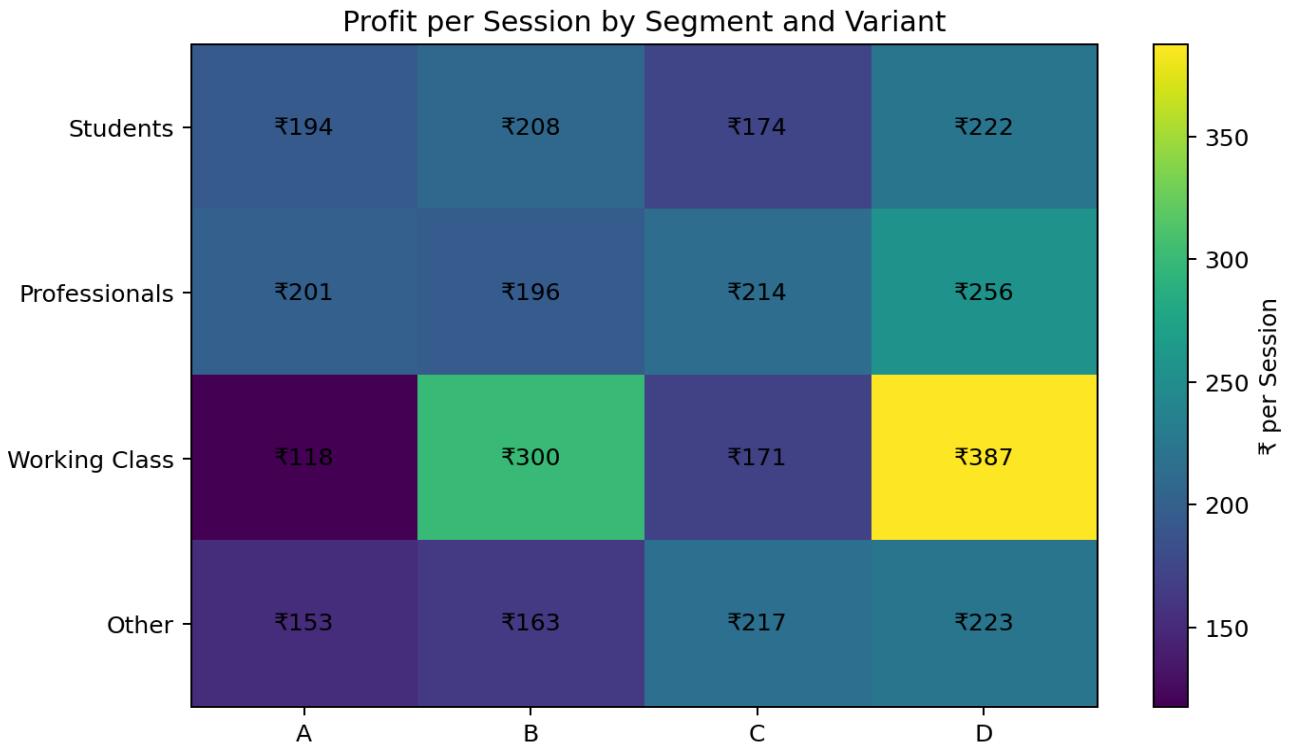
Variant	Sessions	Conversion	AOV	Revenue/session	Profit/session
A - Control	2,000	49.8%	Rs. 631	Rs. 314	Rs. 193
B - Charm Pricing	2,000	53.1%	Rs. 629	Rs. 334	Rs. 205
C - Premium Framing	2,000	45.4%	Rs. 650	Rs. 295	Rs. 181
D - Anchor + Combo	2,000	53.8%	Rs. 682	Rs. 367	Rs. 226

The conversion chi-square test produced chi-square=35.32, p=1.042e-07. The one-way ANOVA for order value produced F=4.02, p=0.007192. Tukey HSD pairwise comparisons are included in the workbook.



6. Customer Segmentation

Consumer activity was mapped into Students, Professionals, Working Class and Other. Segment analysis supports differentiated treatment rather than a universal pricing rule.



The anchor-plus-combo design is the strongest default candidate in the model. Charm pricing remains useful for low-budget users, while premium framing should be tested selectively among high-budget customers.

7. Product Recommendation and Rollout

Advance **D - Anchor + Combo** to a real controlled pilot. It generated the highest modelled profit per session at Rs. 225.83, compared with Rs. 192.64 for control.

Stage 1: expose 10% of eligible traffic for two weeks with persistent user-level assignment. Stage 2: move to 50% only if profit per session remains positive, conversion does not fall by more than two percentage points, and cancellation/refund guardrails remain stable. Stage 3: retain a small holdout after broad rollout.

At 50,000 monthly sessions, the model implies approximately Rs. 1,659,531 incremental monthly contribution profit or Rs. 19,914,369 annually versus control.

A live test is required before claiming business impact because the model does not fully capture operational constraints, competitor reactions, repeat purchase or novelty effects.

8. Resume Positioning

• Designed a four-variant restaurant pricing experiment using anchoring, charm pricing, premium framing and combo architecture to optimize conversion, AOV and contribution profit per session.
• Analyzed 8,000 modelled customer sessions calibrated on genuine UCI restaurant-consumer profiles, applying chi-square tests, ANOVA and Tukey HSD to validate variant-level performance differences.
• Recommended an anchor-plus-combo live pilot, projecting approximately Rs. 1,659,531 incremental monthly contribution profit at 50,000 sessions with segment-specific pricing and experiment guardrails.

Interview wording: describe the uplift as modelled/projected, not achieved in production.